Cyber Risk with Blockchain in Financial Industry

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***Abstract*— Blockchain technology is playing an important role in the digital currency. Every financial institution now has noticed the internet value of the blockchain and in the financial industry especially focus on the risks it could bring to the entire industry. Given the rising number of users, the new emerging technology of blockchain has shown a lot of risks, especially the cyber risks. There are many types of risks under this topic, such as the standard risk, value transfer risk consideration, and smart contract risk considerations. This research will focus on the Smart Contract Risk and Management in the financial industry to discuss the operating model of the risks and list some of the mitigation methods.**

Keywords—blockchain, finance industry, risk consideration, cryptocurrency

I. INTRODUCTION

The financial services are accelerating its adoption of digital technology [1]. Given the consideration of the rising technology of blockchain, and the huge interest on the crypto currency, there are some rapidly rising trends pushing the traditional financial industry putting their eye on the blockchain application. With these new emerging technologies occurred and huge spread in the public, the challenges and risk are also addressed with the unsupervised filed of block application in finance, these major risks have already created some loss in the past, especially the topic of smart contract.

Finance is a broad discipline that examines all types of investments in the commercial sector. The financial sector serves a variety of important roles in society, and financial services are among the economic services it provides. The financial services sector is being impacted by the influence of digital-savvy consumers, the threat posed by large tech corporations, and the changing views of regulators toward new technology [1]. There are numerous forces driving a change in the conventional finance sector.

Blockchain is leading this revolution. As the latest emerging technology, Blockchain was first proposed in 2008 and implemented in 2009. Blockchain has many benefits such as decentralization, persistency, anonymity and auditability. There is a diverse of applications of blockchain technology. The broad range of blockchain applications include social and public services as well as currencies, financial services, risk management, and the internet of things (IoT) [4].

A series of blocks called a blockchain can be thought of as a public ledger where all committed transactions are kept. When more blocks are added to it, this chain keeps expanding [4].

The key features of the blockchain technology include decentralization, persistency, anonymity, and auditability. By combining key technologies like cryptographic hash, digital signature (based on asymmetric cryptography), and distributed consensus mechanism, blockchain may operate in a decentralized environment. Using blockchain technology, a transaction can be carried out independently [4].

As seen in [1, Fig. 1], this is a survey from financial service executives, noticing that these solid data show the blockchain technology has been brought many organizations' attention for multiple reasons, especially in the financial services. Blockchain can therefore significantly reduce costs and increase efficiency [4].

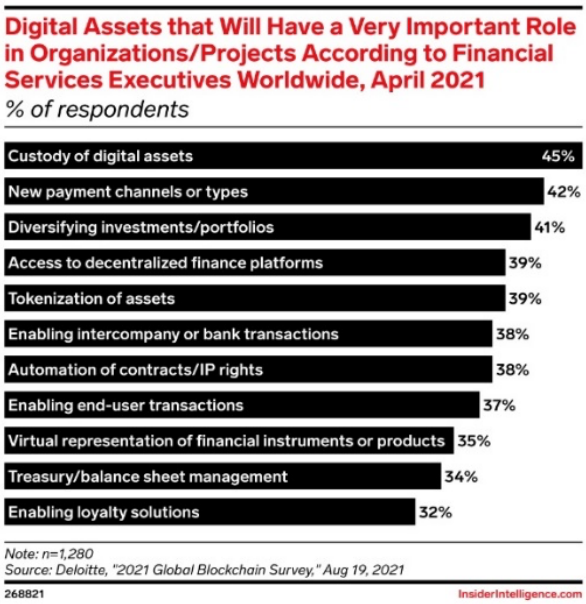


Fig. 1. Financial service executives survey

Compliance and risk are huge in financial industry, and it remains an important role in governing how to apply emerging technology, especially considering the value of the blockchain, now the research topic is discovery some top risks in smart contract and how to mitigate it in financial industry.

II. Review of Financial Industry

The biggest four most prevalent forms of financial institutions in the traditional financial sector are commercial banks, brokerage houses, insurance companies, and investment banks. While modern banking differs from traditional banking in deregulated important facets of the industry, traditional banking is characterized by the application of straight rules [8].

Regulators gave approval for deregulation in order to stimulate growth and improve the quality of the nation's financial services. However, the combination of innovative technologies and the removal of restrictions created complicated cross-border transactions and the rapid movement of capital all over the planet. Deregulation allowed bankers to maximize profits. However, it also created one of the most significant financial meltdowns in human history [8].

Information technology has been innovating and advancing over decades of technical development. Due to these major contributions to the development of computer science and the digitization of information, which encouraged the entire industry entry an new era. Cyborg finance has seen the emergence of a speedier, expanding, intricate, globally integrated business. Financial institutions are all operating in a single financial network [7]. Blockchain technology has developed more and more applications in financial industry. All financial institutions are now highly linked, global financial network that crosses institutions, industries, instruments, and governments, a systemic hazard of being "too linked to fail" arises, like what happened in 2008 and 2020.

When blockchain technology has developed more and more applications in financial industry, the risks of all kinds problems has also rapidly arisen. This is exactly the same process of last decades, the same as the traditional financial industry shifted into modern financial industry, then moved into a cyborg financial, and now it is in the middle of the transition of cyborg financial to decentralized finance.

As the financial system evolves and grows in the past decades, so do its systemic risks. The emergence of cyborg finance has borne two new systemic risks: one related to connectivity "too linked to fail" and the other related to speed that the author terms "too fast to save" [7]. People have witnessed what happened in 2008 and 2020 so far. In recent days the systemic risk of "too big to fail" has drew people' attention again for FTX.

All these problems are because legal change frequently trails technological change. Old laws and old regulations become blunt in the face of sharp, new financial developments. As technological advances transform modern finance into cyborg finance, law's lagging performance has grown more apparent and more consequential [7]. And now the new blockchain technology has made all risks more complexed.

III. Review of Blockchain

After review some financial industry improvement history, and the risks these new emerging technologies brought to the industry. This section will start from introducing what is the blockchain, and what can it brings to some major parties in the financial industry. Then this section will look some major risks and concerns to the different parties.

First of all, a blockchain can be viewed as a sequence of blocks holding a complete list of transaction records, such as the conventional public ledger. In each block, it points to the previous block via a reference that is essentially a hash value of the previous block called parent block [4, Fig. 2].

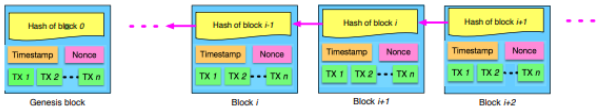


Fig. 2. An example of blockchain which consists of a continuous sequence of blocks

There are two parts inside a block, one is the block header and the other is block body, shown in [4, Fig. 3].

In particular, the block header includes:

- Block version: indicates which set of block validation rules to follow.
- Parent block hash: a 256-bit hash value that points to the previous block.
- Merkle tree root hash: the hash value of all the transactions in the block.
- Timestamp: current timestamp as seconds since 1970-01-01T00:00 UTC.
- nBits: current hashing target in a compact format.

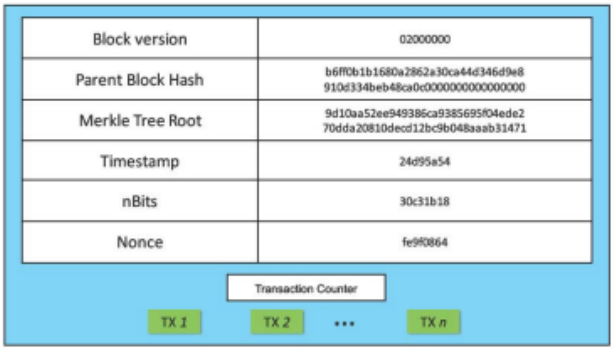


Fig. 3. Block structure

A transaction counter and transactions make up the block body. Depending on the block size and the size of each transaction, a block can contain a maximum number of transactions.

Blockchain verifies the authenticity of transactions via an asymmetric cryptography technique. The signing phase and the verification phase are the two stages of a typical digital signature. The elliptic curve digital signature technique is one of the common digital signature algorithms utilized in blockchains. For transactions in the blockchain, an asymmetric cryptographic digital signature is utilized in an unreliable setting.

Each user will utilize a set of private key and public key to sign documents using the digital signature technique. The transactions are signed using the private key. Public keys that are accessible to everyone in the network are used to access the distributed, digitally signed transactions [4, Fig. 4].

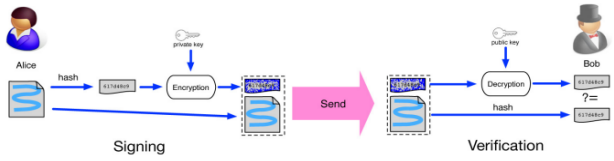


Fig. 4. Digital signature used in blockchain

The key characteristic of blockchain is contributing the benefits in financial industry, which includes decentralization, persistency, anonymity and auditability.

Decentralization is the key characteristic in this situation. A transaction on the blockchain network may be carried out between any two peers (P2P) without the need for central agency authentication. In this manner, blockchain can significantly reduce the server costs and mitigate the performance bottlenecks at the central server [4].

Persistency is vital to the design of the blocks. Each transaction propagating through the network must be verified and recorded in blocks distributed throughout the network. Each block sent is verified by other nodes to validate the transaction. As such, counterfeiting is easy to spot as transactions are very hard to tamper.

Anonymity is the key characteristic to it. It means no central party holding users' personal information. This ensures a certain level of privacy for transactions contained on the blockchain. Note that blockchains cannot guarantee complete privacy protection.

Auditability is important as well. The blockchain allows each transaction to be traced repeatedly to the previous transaction. Accessing any node on the decentralized network can easily review and track previous records. Which improves traceability and transparency of data stored on the blockchain.

Blockchain has the potential to improve the security of privacy and sensitive data. The system can protect the data against these privacy issues, such as data ownership, data transparency and auditability, and fine-grained access control. It can also improve the reliability of security infrastructure.

From [4, Fig. 5], we can see there are a lot of application domains existing in serval industries. The security and privacy, reputation system, and public and social service, Internet of Things and the Finance. Especially with the blockchain application has been applied in multiple fields of industry, the blockchain has shown its potential in all kinds of industry and academia. The financial service has been also provided and addressed to help solving the existing problems.

IV. Top Cyber Risks of Blockchain

Financial risk is a type of danger that can result in the loss of capital to any interested parties. Some more common and distinct financial risks include credit risk, liquidity risk, and operational risk in financial industry, besides that, there are five generic risks to most of the financial institutions: systematic, credit, counterparty or settlement, operational, and legal.

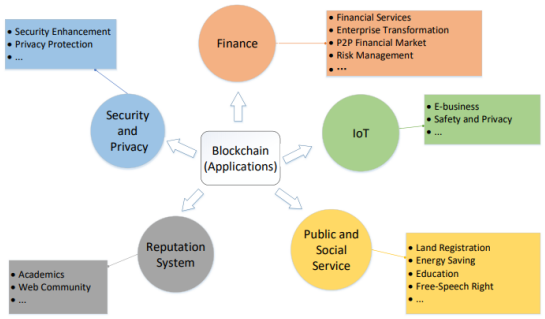


Fig. 5. Representative application domains of blockchain

After the blockchain has joined the revolution, there are some new risks arising t the financial industry, most of them are cyber risks related.

First need to mention is the key management risk, how to secure the digital signature by using the private key is a vital assessment for managing digital asset on blockchain. End users can now select either full control and custody of digital assets or a middle option. Key management risk is the possibility that an end user will not manage their keys, which would result in complete and irreversible loss of those assets [6].

Secondly, risk associated with code and cryptography are also vital to the financial industry. A high level of technical skill, which is currently in short supply, is needed to achieve the required level of assurance. Before, during, and after implementation, blockchain initiatives must test their own code for problems. The risk of employing a poor encryption technique without adding enough randomness to achieve the desired level of security can lead to an exploit. Failing to test the code and rushing to deploy can lead to a substantial loss of money [6].

Consensus and governance risks is slightly overlapping with the legal risk, but converging on a single continuous version of a blockchain ledger is the process of consensus. The practise of routine protocol upkeep and making code modifications is known as governance. Consensus and governance risks include the chance that protocol developers or other responsible stakeholders won't be able to reach an agreement on a timely protocol change. They also include the chance that settlement won't be regarded as a legally binding event because it's possible that a transaction, block of transactions, or the blockchain ledger will eventually be overwritten [6].

After combing the blockchain and financial industry together, introducing common risks to the financial industry, especially acknowledge the top cyber risks of blockchain, the next section will focus on this research object, smart contract in the financial industry, exploring some of the major risk.

V. Smart Contract

Blockchain is ideal for delivering fast and accurate information. It provides immediate, shared and completely transparent information stored on an immutable ledger. A

network can track orders, payments, accounts, production and much more. And because members share a single view of the truth, you can see all details of a transaction end to end. [5] And business runs on the flow of information. The distributed ledger technology help protecting the transaction record immutable and the smart contracts speeds up the transaction. When each transaction occurs, it will be recorded as a block of data, which contains all information about this transaction, and when all these transactions are blocked together in irreversible chain, a blockchain has occurred.

For example, it starts from someone request a transaction, and the transaction will be broadcast to the peer-to-peer network, which is also known as node, and these nodes validates the transaction and the user's status by using the known algorithm, then most of the related transaction information will be involve cryptocurrency, and after verified the transaction, it will combine with other transactions to create a new block of data for ledger. And the new block will be added into the existing blockchain and now it will be permanent and unalterable.

By applying these techniques, it can help people build the trust. They do not need to worry about the transaction details and all of these will be confidential and only disclosure to the related parties and network partners. Transaction will be safe with all these security mechanisms and the efficiency of these transactions will be hugely improved due to the smart contract, which will be executed automatically. [7, Tab. 1] shows the traditional ledgers and blockchain. In smart contracts, remittance is automated and occurs automatically upon fulfilment of the criteria agreed upon and recorded in the code.

TABLE I. COMPARING TRADITIONAL LEDGERS AND BLOCKCHAIN

Ledgers of intermediaries e.g. Bank Account	**Blockchain**
Centralised: Single owner such as a bank makes these ledgers vulnerable since they have a single point of failure that can be hacked.	Distributed: Nobody/everybody owns it because it is distributed across millions of users. It has no single point of failure making it especially difficult and economically unfeasible to hack.
Opaque: Only authorised users can view them or have access to them	Transparent: Anyone can view or access the entire access to them.
Alterable: Errors can be corrected by (internal) users with overriding privileges.	Immutable: Transactions cannot be reversed.
Subject to identity theft: Accounts are often hacked.	Encryption and pseudo-anonymity: This makes it very difficult to hack Blockchain.
Time lag: Can take days or even weeks to complete transactions.	Near-real time: Transactions completed in ten minutes (on average).
Borders: Varying international and conversion fees that can cost up to 20%	Borderless: Same low fees everywhere (usually a few US cents).

[8, Tab. 2] shows the features between the traditional and smart contract. Then it is easy to see the benefits using the smart contract technology based on blockchain in financial industry could bring. Especially when the smart Smart contracts receive the weather data almost in real time and pay the correct party when the specified event happens, the efficacy and the other related benefits are significantly huge.

TABLE II. COMPARING SMART CONTRACTS AND TRADITIONAL CONTRACTS

Smart contracts	**Traditional contracts**
Fast and efficient	Time consuming
Immutable	Mutable by anyone
No need for authority of a third party	Require a third party for authenticity
Cost effective	Expensive
Automatic payment	Manual Payment
Virtual Presence (Digital Signature)	Physical Presence (Wet Signature)

Blockchain is a database maintained and shared by nodes in a peer-to-peer network. Cryptography, distributed storage, a single shared ledger, algorithmic supply and governance of money, programmable money, and permissionless are its characteristics. It's crucial to realize that not all smart contract implementations will be the same perfects as designed, there won't be a single answer for formatting the smart contract to all company in related industry, and a number of their implementations may cooperate to create the new global financial system or in related industry and commerce. The new blockchain risk spectrum appears in this context [6], especially in the smart contract field since it is a programmable contract.

VI. RISKS AND MITIGATION METHODS

In this section, about some monitoring or mitigation handling approaches will be discussed. Despite that technology is a great enable to the financial industry, but it also carries pervasive and potentially high-impact risks to entire industry. After acknowledge the challenges related to IT risk in traditional financial services and the new challenges to the blockchain. How to mitigate and what are the steps to identified top cyber risks is major topics in this research.

Considering the definition of the risk, which refers to the uncertainty and undesirable outcomes. Now, a lot of common risks in financial industry have been discussed, and now considering it from the different viewpoint, what would be the common risks in management, such as there are mainly 4 perspectives about risks category in management. [5, Tab. 3] presents the perspectives of general definition, technology, business and project management. The term risk mainly focusses on cyber risks [5].

TABLE III. RISK DEFINITION

Perspectives	**Definition**
General	A situation involving exposure to danger. The possibility that something unpleasant or unwelcome will happen.
Business	A probability of threat of damage, injury, liability, loss or any other negative occurrence that is caused by external or internal vulnerabilities, and that may be avoided through pre-emptive action.
Technology (ITIL)	A possible event that could cause harm or loss or affect the ability to achieve objectives. A risk is measured by the probability of a threat, the vulnerability of the asset to that threat, and the impact it would have if it occurred. Risk can also be defined as uncertainty of outcome and can be used in the context of measuring the probability of positive outcomes as well as negative outcomes.
Project management	Dual perspective: Overall risk is defined as the effect of uncertainty on the project as a whole. Individual risk is defined as an uncertain event or condition that, if it occurs, has a positive or negative effect on a project's objective.

After reviewing the definition of the risks, it is time to list some of the identified risk parameters. There are 27 perceived risks: access and user right, architecture design, authorizing and provisioning management, business continuity, change management, compliance, consensus protocol, costs, customer experience, data confidentiality, energy consumption, enforcement of contract, governance, integration, interoperability, key management, legal liability, liquidity, performance, privacy, regulatory, reputation, scalability, security, strategy, supplier, and user experience.

There are four major procedures currently in use to assess and manage these risks. These were proposed and are backed by academics, technology consultancies and international organizations.

Since there are some researches which have already summary it up, so we can take the benefit and just review these models and how to use these key functions to manage these risks and what are their benefits [5].

From [5, Tab. 4], for any industry, there are mainly first common step, identification, which includes identifying the current assets and risks. Then, pairing of risk across three policy areas. Apply the considerations and respective control areas across six levels of blockchain for singular risk-based strategy. Identifying the maturity levels and spotting of opportunities for improvement when implementing and using blockchain technology.

TABLE IV. MODELS AND APPROACHES FOR MANAGEMENT OF BLOCKCHAIN-RELATED RISK

Publisher/ Author	Type of Model	Key Functions	Benefits
Cohn Reznick 2018	Singular risk-based strategy	Risk mitigation	Identification of risk considerations and respective control areas across six levels of blockchain
Deloitte 2018	Framework for blockchain risk management	Risk management	Framework for embedding three perceived risk categories - standard, value transfer, smart contract – within business objectives and operations
KPMG 2018	Blockchain Maturity Model	Eight risk identification areas and maturity scoring	Identification of maturity levels and spotting of opportunities for improvement when implementing and using blockchain technology
OECD 2018	Blockchain Primer	Risk identification	Pairing of risk across three policy areas

Secondly, managing the risk, design framework for embedding three perceived risk categories - standard, value transfer, smart contract – within business objectives and operations.

The identification and assessment of hazards constitutes the first of two processes that make up risk management as a whole. Analyzing the assets, identifying threats to those assets, and determining how susceptible each asset is to those threats are all included in this. Both quantitative and/or qualitative methods can be used to evaluate risks. The ongoing management of these risks and the steps taken to reduce them make up the second phase [5].

These are some common procedures when applying the new emerging technology like blockchain. we can see for decentralized finance, which is hugely applied the technology of blockchain, [6, Tab. 5] shows that the popular exchange protocols most are smart contract, but there are others as well.

TABLE V. MOST POPULAR DECENTRALIZED EXCHANGE PROTOCOLS

Protocol name	Protocol type	Price discovery
Ox	Exchange	Off-chain order books
(Air)Swap	P2P/OTC	P2P negotiation
Bancor	CFMM	Smart contract
Balancer	CFMM	Smart contract
Curve	CFMM	Smart contract
Kyber Network	Reserve aggregator	Proposal by maker
UniSwap	CFMM	Smart contract

[a.] NOTE: CFMM, constant function market maker.

Common standards for the entire industry and country can help ensure that all kinds of products and services are safe and secure from cyber-attacks. The US National Institute of Standards and Technology (NIST) has developed a framework for this purpose, although its relevance is not immediately clear. but it is important to have a regulation or industry standard for different kinds of products.

The World Economic Forum's 2019 report on value creation with blockchain technology highlights the need for a new analytical framework to assess and manage the risks associated with the technology. For example, interoperability is a key risk area as connecting different types of blockchains for transactions and commerce introduces new risks and barriers [5].

Some of the risks and challenges of using blockchain sustainably are considered standard across industries and use cases, while others are based on specific applications of the technology. Understanding how decision makers perceive the risk of transitioning to a distributed ledger is at the heart of the risk management process. Well-planned governance policies must be developed to maximize the benefits of blockchain technology and ensure its sustainable use at all lifecycle stages. An analysis of which legal framework conditions are possible may support further research on risk management for the future of the technology.

Above is the general governance outside the company, which is mainly about building the compliance for the entire industry and briefly mentioned how to identify, match and pair the risk and their duration and maturity, and how to mitigate the risk inside any organization is the other focus point of this research. According to others' research with test case and

conclusion. [6, Fig. 6] shows a detailed list of how to maintain control within the entire lifecycle of blockchain.

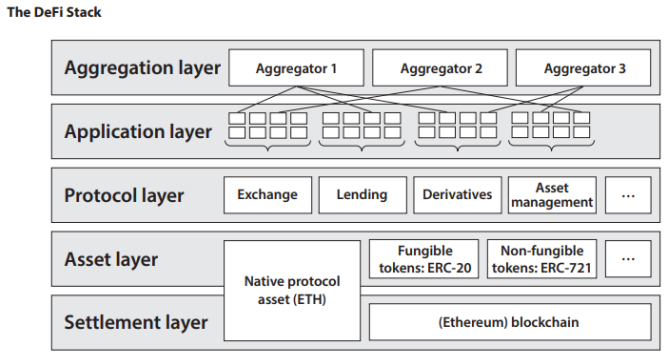


Fig. 6. The DeFi stack

From pre-implementation, the company need to do assessment and have consensus on senior management, and convinces vendors, then make some proper framework. During the governance, it has to make some plans, such as business continuity plan, disaster recovery plan and periodically review the risk to ensure their plan is effective.

When during the development, make sure providing enough training for developers and vendors' stuff. Document every requirement for business and list support plan. After the implementation, make sure the test function and audit teams are effective and when necessary, the two audit teams are better in reviewing the code. Remember that designing and implementation are two different functions and make sure they cooperate tightly. In the end, make sure the operational procedures are managed.

The security functions are provided in the process for manage loss, theft of private keys, or at least have a standard process. Then if there are enough resources, developing for endpoint device security and network vulnerabilities are recommended. Also, noticing that when any vulnerabilities are found, report to other related parties and stakeholders are mandatory.

In the end, noticing the transactions may need monitor before, in use, and after the transaction to make sure the transaction fees are budgeted appropriately and the entire transaction are immutable and traceable.

These are recommended control description during the entire life-cycle, from pre-implementation, governance, consensus to development, security and transaction. These control methods are tested among many other controls and chosen as most effective, here this research paper based on that information giving a detailed procedure about how to design the control mechanism during the entire lifecycle. This can be used for protecting the use of blockchain in financial industry, specifically the smart contract is workable.

In conclusion, when developing blockchain related emerging technology, the company should not only focus on the industry compliance, more important is how to governance inside the company, have a certain procedure and process for dealing any cases is this section focus point, and listing the check point and detailed control methods listed above are considered with many cases has been proved as successful.

TABLE VI. INTERNAL BLOCKCHAIN DEVELOPMENT PROCEDURE

	BLOCKCHAIN CONTROL DESCRIPTION
Pre-implementation	1. The enterprise has created and maintains a blockchain technology business case assessment. 2. Senior management supports deployment of blockchain technology. 3. A governance framework for blockchain technology has been created and approved. 4. Vendors are properly vetted by the enterprise.
Governance	1. Management oversight is periodically reviewed to ensure that the governance framework for blockchain is effective. 2. The enterprise includes regulatory risk in its risk assessment of blockchain technology and periodically reviews the assessment to maintain relevance. 3. The enterprise has a business continuity plan for the blockchain solution. 4. The enterprise has a process for managing blockchain technology vendors.
Development	1. The enterprise provides adequate blockchain training for existing developers. 2. Business requirements for the blockchain solution have been documented and approved by the appropriate person/group within the enterprise. 3. The blockchain solution is adequately designed to support business requirements (e.g., platform architecture is consistent with enterprise needs). 4. The enterprise has a test strategy/test plan for the blockchain solution. 5. The enterprise has a plan for deploying the blockchain solution. 6. Features for the blockchain solution have been adequately deployed. 7. The enterprise has designed and implemented standard methods and procedures for operational. 8. The enterprise has a blockchain change management program that operates effectively.
Security	1. The enterprise has implemented a process for managing loss or theft of private keys. 2. A process is in place to manage blockchain network vulnerabilities. 3. A process exists to manage endpoint security for devices using the blockchain solution. 4. Vulnerabilities identified during source-code reviews are properly managed in terms of mitigation, action plans and communication to relevant stakeholders.
Transaction	1. A process ensures that transactions on a blockchain are immutable and traceable. 2. Transaction fees are monitored. 3. Transaction fees are budgeted appropriately.
Consensus	1. The enterprise has developed and implemented consensus functionality on the relevant protocols. 2. The enterprise has designed and implemented the necessary infrastructure to support blockchain mining. 3. Infrastructure for cloud-based/leased mining is appropriate.

VII. CONCLUSION

Regulating the emerging, new financial industry will be one of the most challenging endeavors for policymakers in the coming years. It is understood that much of the difficulties of financial regulation lie in the actual drafting, passage, implementation, execution, and enforcement of new rules and regulations [7].

Proposed principles of regulatory design for policymakers as they contemplate fresh rules and regulations for blockchain application. Some of the proposed principles can be perceived as competing, complementary, and crosscutting. These principles are intended to serve as guideposts and not roadblocks for creating a workable framework for the new financial industry.

This Article offers an early systemic account of this complex, ongoing metamorphosis and its wide-ranging policy ramifications for financial regulation. It identifies particular dangers, systemic risks, and current regulatory shortcomings [7].

In the end, it is intended to serve as an early framework for further study on how best to regulate the emerging, new financial industry with these applications, such as smart contract [7]. Other than the industry regulatory and compliance, the inside governing is even more important as well. With reviewing the financial industry and blockchain technology, and the better understanding of smart contract, go over multiple cases for the risk and mitigation method, these methods are considering as useful in governing the risk with the development of the industry, and these control methods covers the entire lifecycle from design to the implement and after care, which should be enough for now.

VIII. Future Work

In terms of future research paths, incentives for doing research into the needs and risk management for standards-based blockchain.

For the long-term usage of blockchain, a standards framework and a focused discussion of requirements management have not yet been developed. Identifying new kinds of leadership and consulting might be made possible by filling the gaps in both the research and management fields [3].

With the conclusion that considering risk and requirement parameters in accordance with the general understanding of risk and the four previously defined viewpoints—general, business, technology, and project management—could be beneficial for risk management within various blockchain environments. Based on a high level of interest across several industries and use cases, suggest that a good approach to risk management across the whole adoption process is essential for the long-term adoption and usage of blockchain technology [3].

Developing an integrated, normative framework for operationalizing risk factors as well as a tool for more effectively assessing and managing these individual risks as a tangible outcome of the future work. Then, depending on the results, the framework might be adjusted and tested across sectors and use cases before being further developed towards market maturity [3].

DeFi employs an architecture with several layers. Every layer serves a certain function. Everybody can build on, rehash, or utilize other components of the stack thanks to the layers' ability to build on one another and form a highly extensible architecture. The hierarchical nature of these levels must also be understood: They can only be as safe as the layers beneath them. All further levels wouldn't be safe, for instance, if the blockchain at the settlement layer were to be hacked. Similar to this, all attempts at decentralization on succeeding levels would be futile if using a permissioned ledger as the base [3].

This new emerging structure of DeFi could be the exciting new world of financial industry with the application of blockchain.

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